



Mentamind – AI-Powered Healthcare & Employee Wellness Platform



Department of Computer Science & Engineering
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Submitted By :

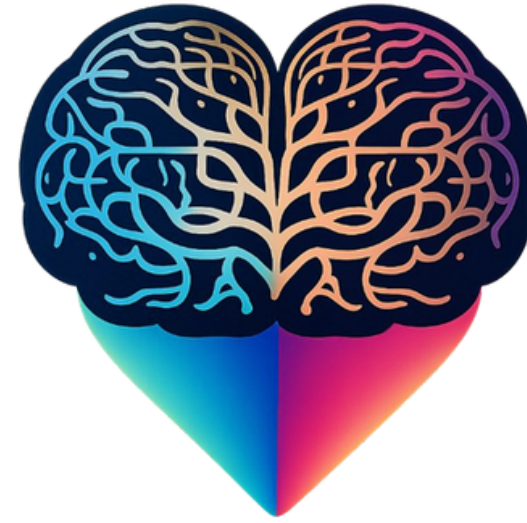
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MentaMind

Your trusted companion for mental well-being.



Agenda

- Introduction
- Motivation
- Objective & Problem Statement
- Literature Survey
- Methodology
- Implementation
- Result
- Conclusion
- Future Scope
- References



Introduction

- Mental health challenges like stress, anxiety, burnout, and depression are rapidly increasing among students and professionals.
- Traditional therapy faces barriers such as high costs, long waiting times, limited accessibility, and social stigma.
- AI and cloud technologies enable scalable, affordable, and accessible mental healthcare solutions through digital platforms.
- Mentamind integrates AI chatbot support, therapist booking, mood tracking, wellness resources, and organisational analytics in one system.
- Built using modern cloud-native technologies like Next.js, Node.js, NestJS, MongoDB, Redis, and Amazon Web Services to ensure scalability, security, and reliable deployment.

Increasing cases of stress,
anxiety, burnout, and
depression

Social stigma prevents many
users from seeking help
openly

Need for an integrated and
accessible digital wellness
ecosystem



Motivation

Cost & geographic
accessibility

Expensive and
inaccessible in
rural populations
with limited
connectivity.



- Provide 24×7 AI-based emotional support and therapist connectivity
- Enable mood tracking and emotional wellness analytics
- Support employee wellness monitoring for organizations
- Deliver a secure, scalable, and multilingual platform



Existing mental healthcare solutions lack an integrated, affordable, and accessible platform that combines AI emotional support, human therapist connectivity, wellness analytics, and organizational wellness management.



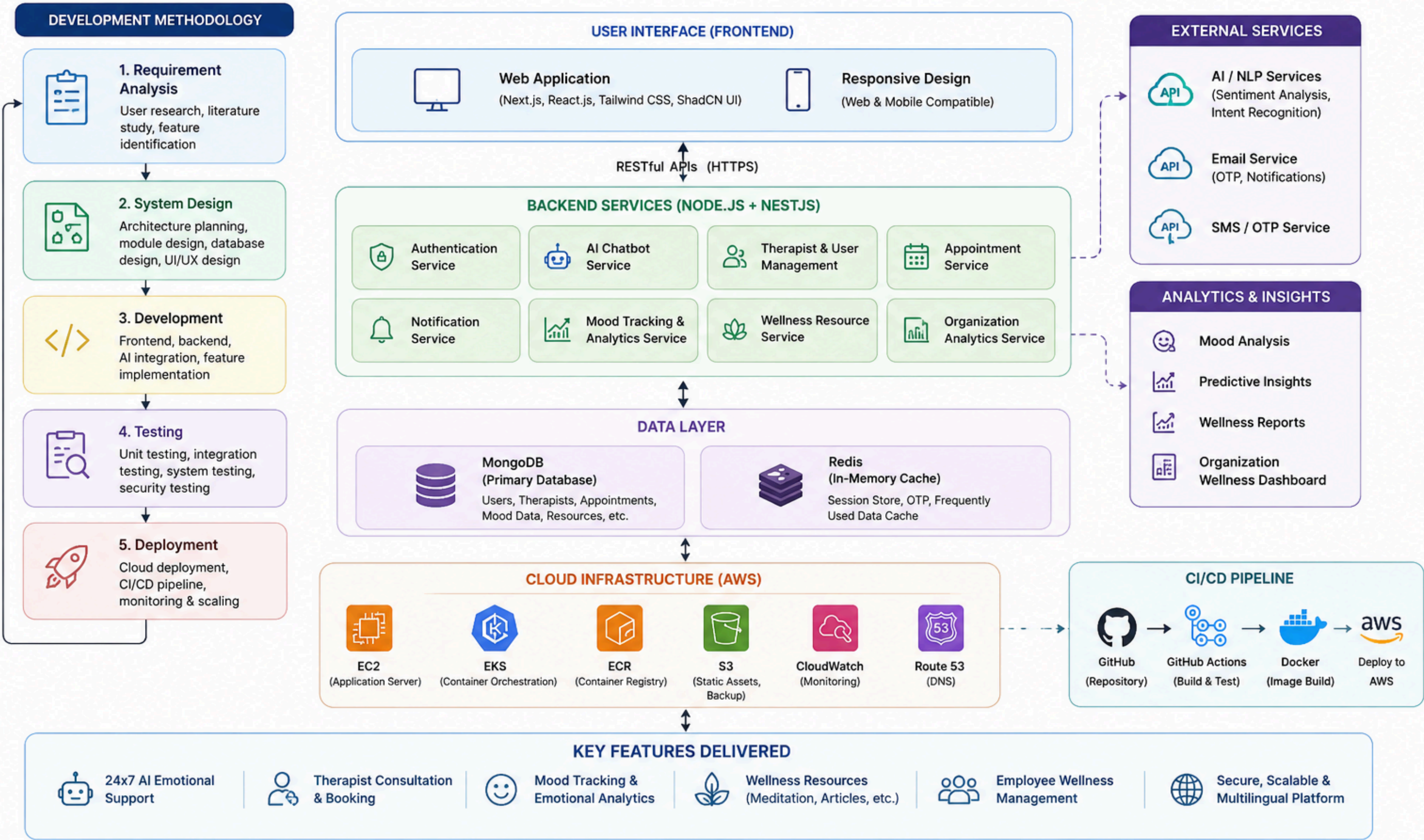
Objective & Problem Statement

Literature Survey

- Major research by S. Abd-Alrazaq on AI-powered CBT chatbots and Timothy Bickmore on conversational AI highlights the growing role of AI in digital mental healthcare
- Studies from Harvard Business Review show strong links between employee wellness, productivity, and organizational performance
- Analysis of online therapy platforms such as BetterHelp, Talkspace, and YourDOST for remote therapist accessibility
- Review of wellness and meditation applications including Calm and Medito for mindfulness and stress management
- Study of AI emotional support platforms like Wysa and Woebot Health reveals the research gap for an integrated AI + therapist + analytics ecosystem

METHODOLOGY

Mentamind – AI Powered Healthcare & Employee Wellness Platform



IMPLEMENTATION

Mentamind – AI Powered Healthcare & Employee Wellness Platform



FRONTEND IMPLEMENTATION

- Built with Next.js, React.js and TypeScript
- Responsive UI using Tailwind CSS & ShadCN UI
- User-friendly interfaces for chatbot, dashboards, bookings & resources



BACKEND IMPLEMENTATION

- Developed using Node.js and NestJS
- Modular services for authentication, chatbot, appointments, analytics & notifications
- RESTful APIs secured with JWT



DATABASE & STORAGE

- MongoDB for storing user, therapist, mood, and appointment data
- Redis for caching, session management & OTP storage
- Optimized indexes for high performance



AI & ANALYTICS IMPLEMENTATION

- NLP-based chatbot for emotional support
- Sentiment analysis & intent recognition
- Mood tracking, trend analysis & predictive insights
- Personalized wellness recommendations

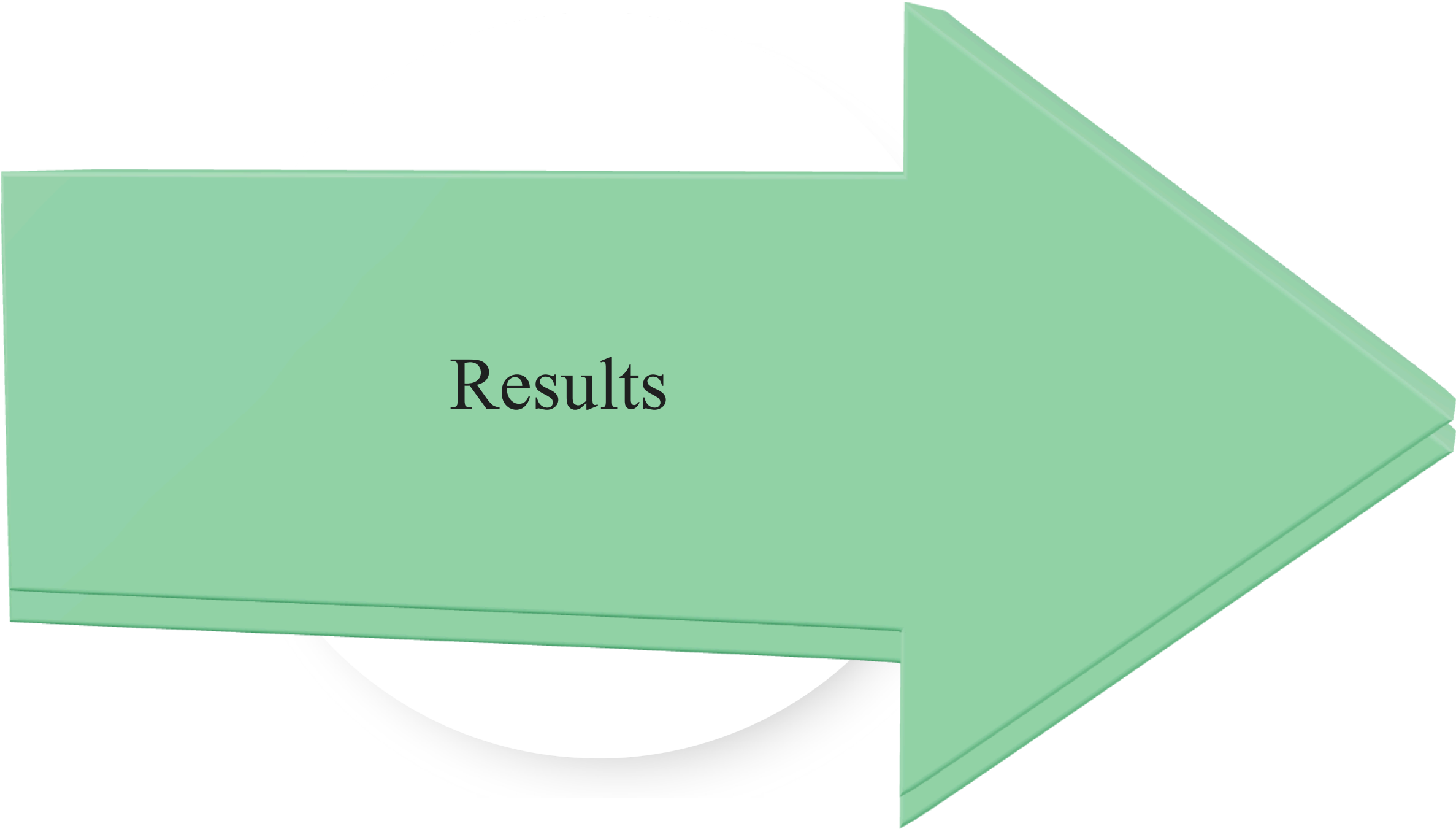


DEPLOYMENT & SECURITY

- Deployed on AWS cloud infrastructure
- Dockerized services with CI/CD pipeline
- Data secured using HTTPS, JWT, BCrypt & RBAC
- Scalable, reliable & highly available system

SYSTEM ARCHITECTURE





Results



Mood and Symptom Tracking

Personalized insights for early detection.



MentaMind App

Your personal mental health companion.



Multilingual Support

Breaking barriers with Indian regional languages.



Therapist Marketplace

Book online sessions with certified psychologists.



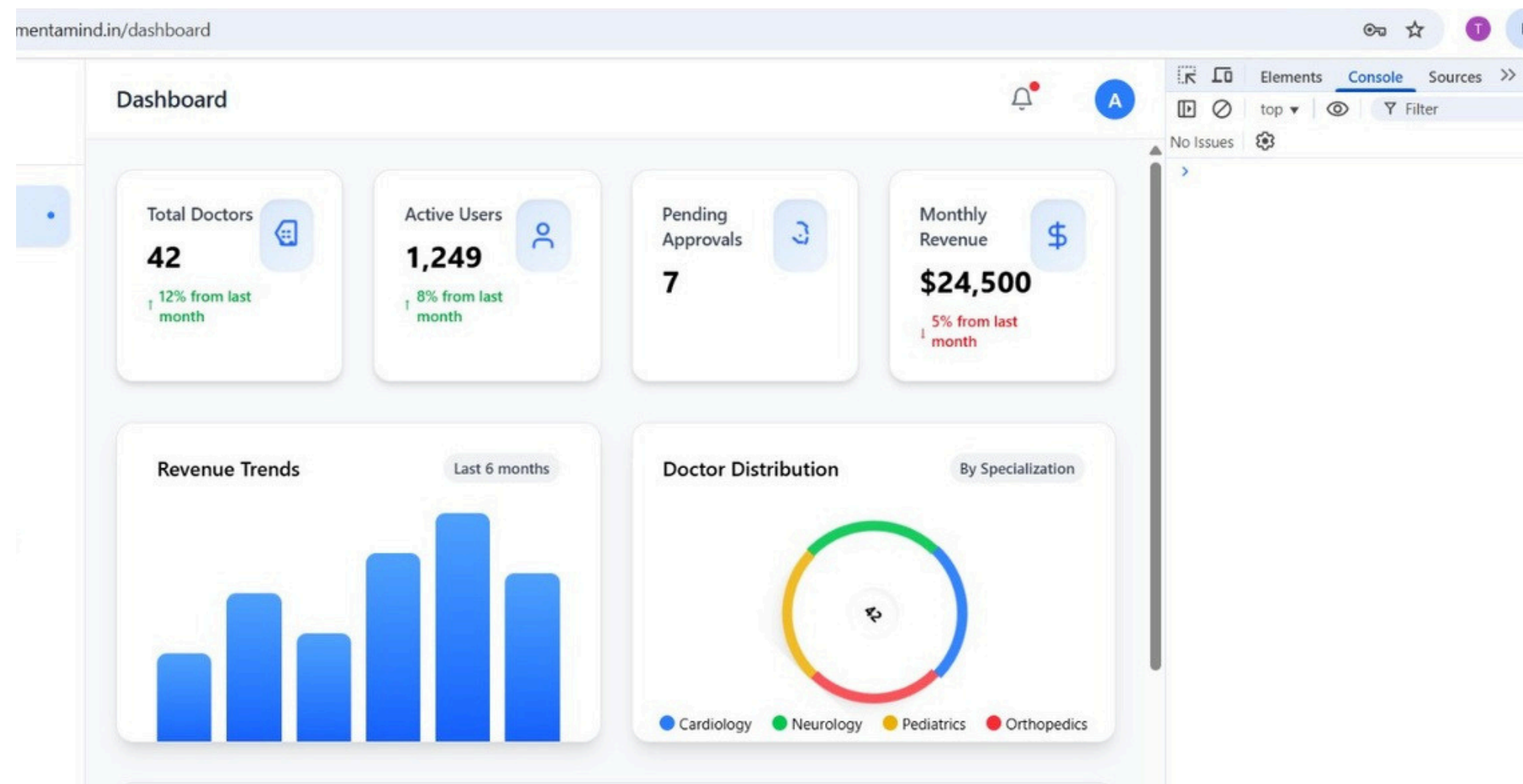
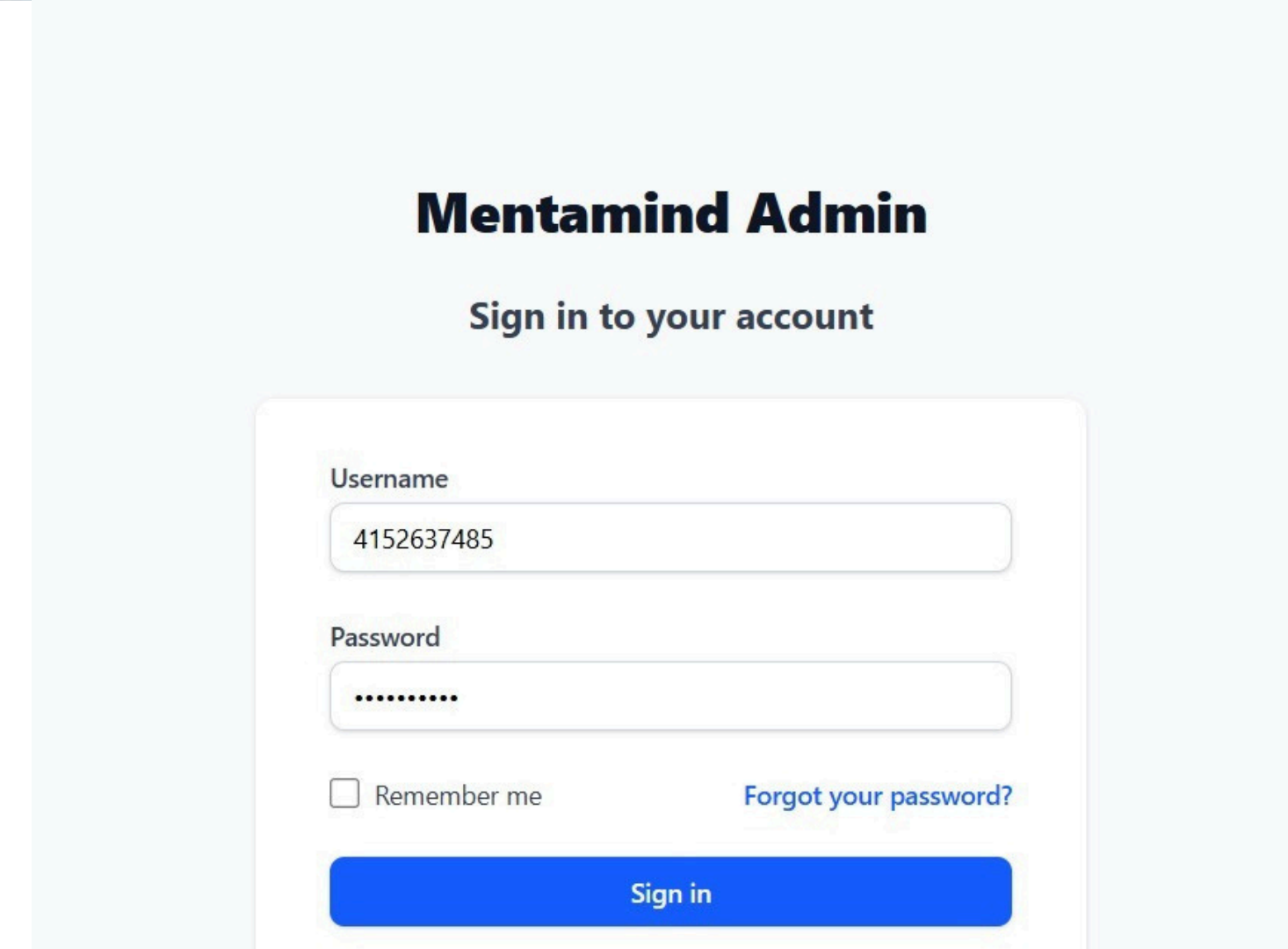
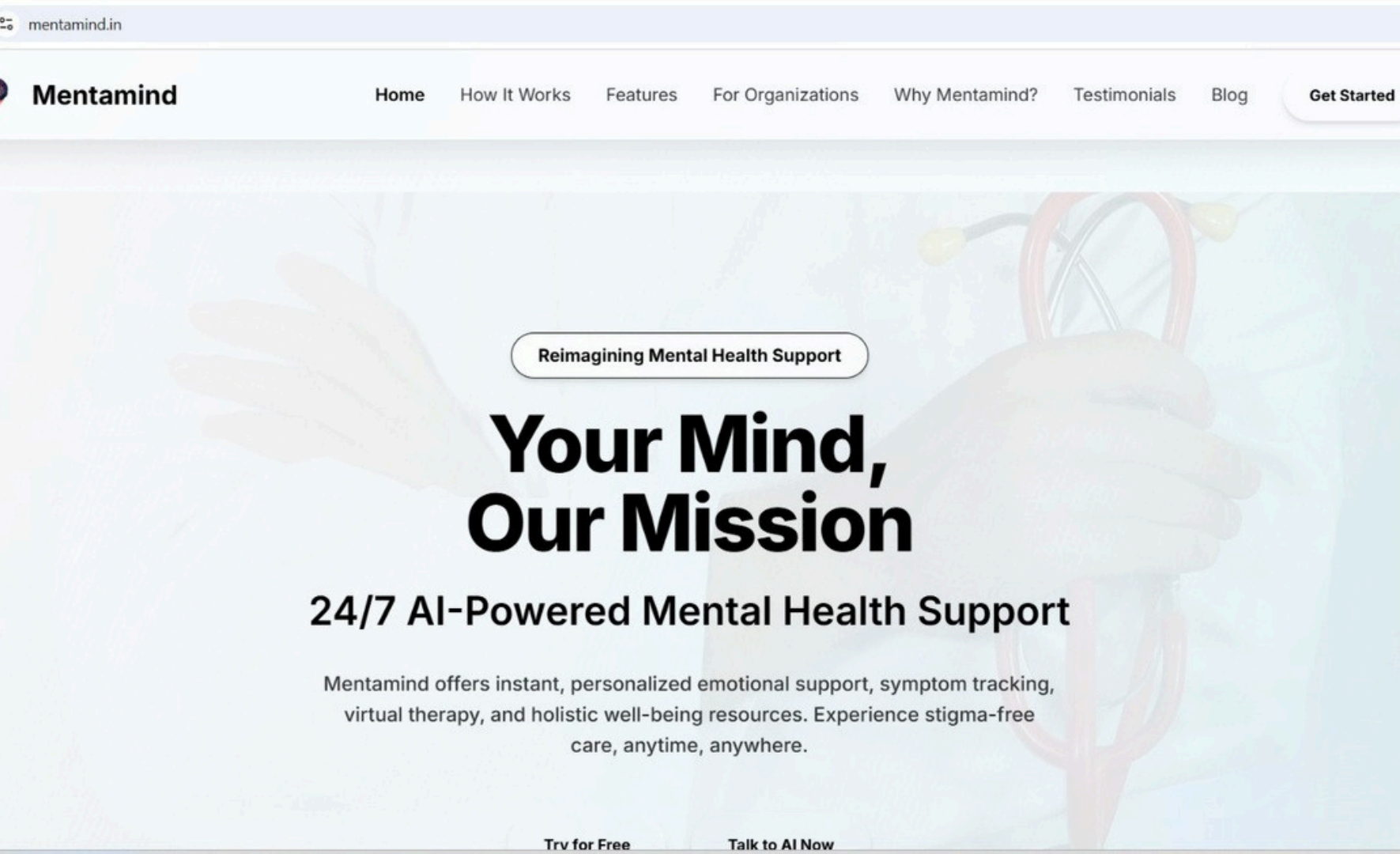
Wellness Resources

Guided meditation, stress relief exercises, and lifestyle coaching.



AI Chatbot

24/7 emotional support, symptom check, and self-help guidance.



OUR ADVANTAGE

	Mentamind	Wysa	YourDOST	Practo	InnerHour	1mg
AI Symptom Diagnosis	✓	✗	✗	✗	✗	✗
24/7 Conversational AI	✓	✓	✗	✗	✗	✗
Teleconsultation	✓	✗	✓	✓	✓	✓
Guided Meditation	✓	✗	✗	✗	✓	✗
Online Pharma	✓	✗	✗	✓	✗	✓
Multilingual Support	✓	✗	✗	✗	✓	✗
Data Security	✓	✗	✗	✗	✗	✗

Conclusion



Integrated Platform

Combines AI support, therapist access, analytics, and wellness resources in one platform.



Data-Driven Insights

Mood tracking and analytics provide personalized emotional insights.



Secure & Scalable

Built with robust security and cloud infrastructure for reliable and scalable access.



Accessible Support

24x7 AI assistance and easy therapist access make mental healthcare more inclusive.



Organizational Wellness

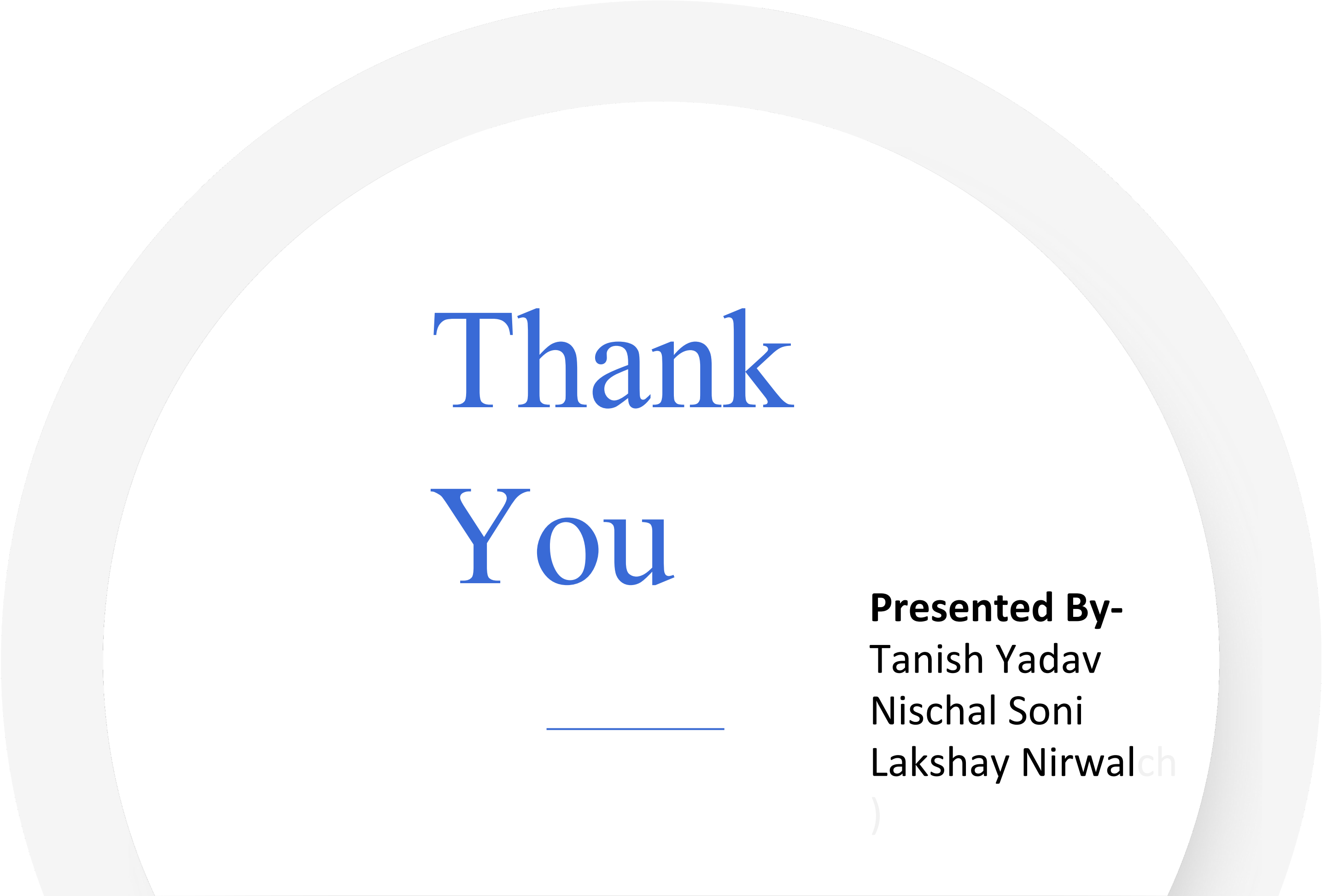
Empowers organizations with analytics to monitor and improve employee well-being.

FUTURE SCOPE

- **Voice-Based Support:** Integrate speech recognition and voice tone analysis for deeper emotional understanding.
- **Facial Emotion Recognition:** Use computer vision to detect real-time emotions for more accurate insights.
- **Wearable Integration:** Sync with wearables to monitor physiological signals for real-time wellness tracking.
- **Advanced AI Models:** Incorporate deep learning for adaptive therapy pathways and nuanced emotional support.
- **Mobile & Video Consultations:** Develop native mobile apps and secure video consultation for enhanced accessibility.
- **Multilingual Expansion:** Support more languages to reach wider and diverse populations.
- **Enterprise & Government Expansion:** Scale solutions for large enterprises, institutions, and public healthcare systems.
- **Blockchain Integration:** Implement blockchain for secure, immutable, and privacy-focused record management.

References

1. S. Abd-Alrazaq et al., “Artificial intelligence–powered cognitive behavioral therapy chatbots: A systematic review,” *International Journal of Mental Health Systems*, vol. 18, no. 1, 2024.
2. T. Bickmore et al., “Conversational AI for mental health,” *NPJ Digital Medicine*, vol. 4, no. 1, 2021.
3. Harvard Business Review, “Employee wellness and workplace productivity,” *Harvard Business Review*, 2022.
4. BetterHelp, “Online Therapy Platform,” BetterHelp Inc. [Online]. Available: <https://www.betterhelp.com>. Accessed: 2025.
5. Talkspace, “Online Therapy and Mental Health Services,” Talkspace Inc. [Online]. Available: <https://www.talkspace.com>. Accessed: 2025.
6. Calm, “Meditation, Sleep, and Relaxation Application,” Calm.com Inc. [Online]. Available: <https://www.calm.com>. Accessed: 2025.
7. Wysa Ltd., “Wysa: AI-Based Emotional Support Chatbot,” Wysa Ltd. [Online]. Available: <https://www.wysa.com>. Accessed: 2025.
8. Woebot Health, “Woebot: AI Mental Health Companion,” Woebot Health Inc. [Online]. Available:



Thank You

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